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Revolutionizing Lower Completions Design Through Unsupervised Machine Learning Clustering: An Integration of Well Construction Data and Reservoir Characteristics

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Abstract

This paper presents a novel approach that utilizes unsupervised machine learning clustering (UMLC) to enhance the completions process in reservoir management. By integrating real-time data streams from multiple sources, this method harmonizes and transforms data into a unified format for seamless cross-disciplinary analysis. The goal is to optimize zone selection for completions design, enabling better decision-making in a shorter timeframe and avoiding data pitfalls between Well Construction and Completions stages. Additionally, this paper compares the zones identified using the same wells from previous studies.

The proposed integration focuses on the ingestion and processing of real-time data from various sources. This data undergoes harmonization and transformation into a standardized format, which allows for seamless integration and analysis through sophisticated UMLC algorithms that delineate zones based on the data characteristics throughout the lateral well section. Subsequently, these zones are assessed by interpreters to pinpoint defined potential pay zones, non-productive intervals, and optimal locations for completions design equipment, emphasizing production and isolation. This analytical framework fosters a collaborative decision-making environment that promotes synergy among multiple disciplines, including geology, petrophysics, reservoir engineering, and completions, ensuring that all pertinent insights are incorporated into the completions process.

The comparative evaluation between interpreter-generated results and UMLC outcomes highlights notable discrepancies in zonation accuracy. The data-driven clustering results yield a more precise and nuanced assessment of zones, enabling interpreters to distinguish between high-quality and low-quality productive zones, as well as non-productive zones. This enhanced accuracy is attributed to the incorporation of data science and machine learning methodologies, which provide users with a deeper understanding of the well and reservoir characteristics. Additionally, the methodology includes a quality control mechanism to establish the feature importance scores of the clustering and to compare the analytical drivers.

The amalgamation of UMLC with conventional reservoir mapping techniques signifies a substantial leap forward in reservoir management practices. The data-centric clustering outcomes deliver a degree of granularity and precision that bolsters interpretation, ultimately yielding optimal zonation and extended

productivity of lateral wells. By harnessing the capabilities of data science and machine learning, this approach ensures more cost-effective and timely completions processes.

Introduction

The optimization of lower completions in horizontal wells is a critical component of modern reservoir management, particularly in heterogeneous clastic and carbonate formations. Traditional completions design relies heavily on deterministic interpretations of petrophysical logs and geophysical inversions, often requiring manual integration of data from multiple disciplines. While effective in many cases, these methods are time-consuming, subject to interpreter bias, and may overlook subtle but significant variations in reservoir quality. Recent advances in machine learning, particularly unsupervised clustering techniques, offer a promising alternative. These methods can automatically identify patterns and groupings in high-dimensional datasets without requiring labeled training data. When applied to petrophysical and geophysical logs, unsupervised clustering can reveal natural zonation patterns that correspond to variations in porosity, permeability, saturation, and other key reservoir properties.

Recent advancements in real-time data acquisition and digital infrastructure have enabled the continuous ingestion of diverse data streams during drilling operations. These include logging-while-drilling (LWD) measurements, mud logging, drilling mechanics, and borehole imaging. However, the integration of these heterogeneous datasets into a unified analytical framework remains a significant challenge. Differences in data formats, sampling rates, and spatial resolution can hinder seamless analysis and delay critical decisions.

To address these limitations, this study introduces a novel approach that leverages unsupervised machine learning clustering (UMLC) to process and analyze real-time data streams for completions optimization. The proposed workflow focuses on harmonizing and transforming multi-source data into a standardized format suitable for clustering analysis. By applying UMLC algorithms to this unified dataset, the method delineates reservoir zones based on intrinsic data patterns rather than predefined thresholds or interpreter assumptions.

This approach enables the identification of high-quality productive zones, low-quality intervals, and non-productive sections in near real-time, supporting more agile and informed completions design. Furthermore, the clustering results are compared with interpreter-defined zonation from previous studies on the same wells, providing a basis for evaluating the accuracy, consistency, and added value of the machine learning approach.

The integration of UMLC into the completions workflow also fosters a collaborative decision-making environment. By providing interpretable clustering outputs and feature importance metrics, the method facilitates communication among geologists, petrophysicists, reservoir engineers, and completions specialists. This ensures that all relevant insights are incorporated into the final design, reducing the risk of suboptimal completions and enhancing overall reservoir performance.

Unsupervised Clustering in Oil & Gas Applications

Unsupervised clustering has emerged as a transformative analytical approach in the oil and gas industry, particularly in the domains of reservoir characterization, well log interpretation, lithofacies classification, and completions optimization. Unlike supervised learning, which relies on labeled datasets, unsupervised clustering algorithms autonomously identify patterns and groupings within data based solely on intrinsic similarities. This capability is especially valuable in subsurface environments where labeled data is scarce, ambiguous, or costly to obtain.

In the context of reservoir engineering and geoscience, clustering techniques are employed to:

- Segment well logs into lithological or petrophysical facies.

- Identify flow units or reservoir compartments.
- Detect anomalies or outliers in production or drilling data.
- Support completions design by delineating zones with similar reservoir behavior.

These applications benefit from clustering's ability to reduce dimensionality, enhance interpretability, and reveal hidden structures in complex, multivariate datasets.

Case Studies and Published Applications

Numerous studies have demonstrated the efficacy of unsupervised clustering in oil and gas. Unsupervised learning has emerged as a powerful tool in subsurface data analysis, enabling geoscientists and engineers to uncover hidden patterns, classify geological features, and optimize reservoir development strategies without the need for labeled datasets. Across a wide range of applications—from facies classification and formation testing to hydraulic fracturing design and reservoir pressure modeling—unsupervised learning techniques such as clustering, dimensionality reduction, and anomaly detection have demonstrated significant value.

The application of unsupervised clustering significantly enhanced the detection and mapping of paleochannels in this study. By leveraging curvature-based seismic attributes—particularly the positive curvature—the researchers applied clustering algorithms such as Fuzzy C-means and BIRCH to segment the seismic data into meaningful geological structures. These methods enabled the objective identification of paleochannels without the need for labeled training data, thereby reducing manual interpretation bias and increasing the reproducibility of results. The use of Principal Component Analysis (PCA) further refined the clustering output by reducing dimensionality and enhancing the contrast between paleochannels and surrounding formations. This automated, data-driven approach not only improved the accuracy of paleochannel delineation but also demonstrated scalability and adaptability across different geological contexts, offering a robust framework for future seismic interpretation tasks (Hungund & Jiang, 2024).

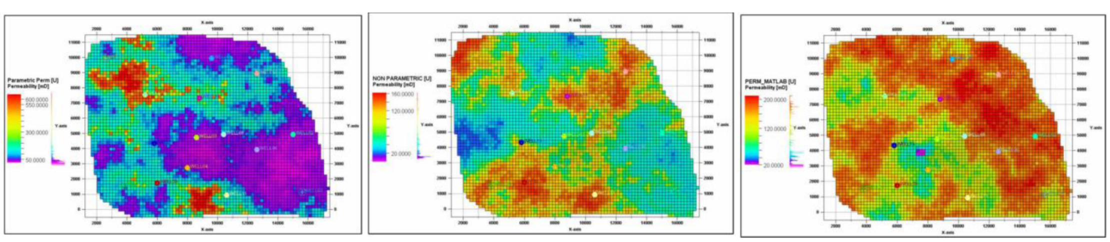


Figure 1—Parametric, Non-Parametric and Machine Learning permeability maps. As per author, Machine Learning results shows more uniform permeability than the two other methods and clearly shows two different hydraulic flow units in the upper reservoirs with high-permeability section in dark red and medium-permeability sector in yellow-green color.

Unsupervised clustering played a foundational role in enhancing reservoir characterization in this study. Specifically, the k-means clustering algorithm was employed to classify well log data into distinct hydraulic flow units based on similar petrophysical properties. This unsupervised approach enabled the identification of electrofacies without relying on pre-labeled data, allowing for a more objective and data-driven segmentation of the reservoir. The delineation of these flow units was critical for improving the accuracy of permeability predictions, which were subsequently used in machine learning models. Among the tested methods, the machine learning approach incorporating k-means clustering yielded the lowest mean absolute error (MAE) and mean squared error (MSE) when validated against core data. Furthermore, permeability models derived from k-means clustering demonstrated superior performance in history matching simulations, achieving the least deviation from observed production and pressure data. These findings underscore the value of unsupervised clustering in capturing reservoir heterogeneity and enhancing the predictive capabilities of machine learning workflows in subsurface modeling (Koray et al., 2023).

In addition, the machine learning method showed the least error calculation using MAE and MSE.

Table 1—Different Permeability Calculation Methods

Permeability calculation method	Error in cumulative oil	Error in cumulative water
Parametric method	-0.70%	14.60%
Non-parametric	-1.20%	15.50%
Machine learning using k-means clustering method	-0.30%	4.20%

Unsupervised learning was central to the predictive maintenance strategy developed in this study for Electrical Submersible Pumps (ESPs) in water supply wells. Due to the limited availability of labeled real-time data—particularly the absence of downhole pressure and temperature readings—the researchers adopted an unsupervised anomaly detection approach using the Isolation Forest algorithm. This method enabled the identification of abnormal patterns in sensor data such as motor current, discharge pressure, and flow rate, which were indicative of impending ESP failures. By clustering data based on hidden structures and detecting deviations from normal operational behavior, the model could predict failures 90 to 120 days in advance. These early warnings allowed for proactive workover planning, significantly improving well availability and water injection capacity. The unsupervised model, combined with statistical metrics like Mean Time Between Failures (MTBF), achieved a 70% success rate in blind tests and demonstrated the practical value of anomaly-based failure prediction in data-constrained environments (Reddicharla et al., 2023).

Although the primary focus of this study was on supervised learning models, unsupervised learning played a foundational role in the early stages of model development. Initially, the authors explored unsupervised clustering techniques, including t-distributed stochastic neighbor embedding (t-SNE), to identify patterns in the injection/falloff test data. However, these methods lacked the predictive accuracy required for treatment design, prompting a shift toward supervised classification models. The unsupervised phase was instrumental in understanding the structure of the dataset and guiding feature selection, which ultimately improved the performance of the supervised models. This transition highlights the complementary role of unsupervised learning in preprocessing and exploratory data analysis, especially when dealing with complex, high-dimensional datasets in hydraulic fracturing applications (Khan et al., 2022).

Unsupervised learning played a foundational role in this study's innovative approach to diagnosing and managing water encroachment in heterogeneous oil reservoirs. The methodology began with the application of time-series clustering techniques to group wells based on their water cut behavior over time. This unsupervised phase was critical for uncovering hidden patterns in dynamic production and injection data —such as water cut, water-oil ratio (WOR), bottom-hole pressure, and injection rates—without relying on predefined labels or assumptions.

The clustering was performed using Dynamic Time Warping (DTW), a technique well-suited for aligning time series of varying lengths and temporal distortions. This allowed the researchers to identify families of wells exhibiting similar water cut trends, both temporally and spatially. These clusters served as the basis for defining "type curves" that represent average water cut behavior within each group, enabling a standardized analysis of water influx dynamics across the field.

Once the clusters were established, they were used as target labels in a supervised learning phase, effectively transforming the unsupervised insights into a structured classification problem. A decision tree model (CART) was then trained using static geological features—such as porosity, permeability, rock type, and fracture properties—as inputs. This allowed the team to quantify the relative importance of each static variable in influencing water encroachment behavior across different well clusters.

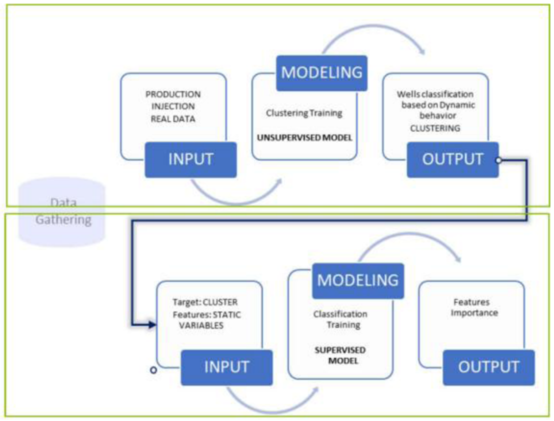


Figure 2—Unsupervised to Supervised Machine Learning method workflow

The integration of unsupervised and supervised learning provided a powerful framework for linking dynamic production behavior with underlying geological characteristics. It enabled the generation of water saturation maps and predictive models that not only visualized current water movement but also forecasted future encroachment trends. This hybrid approach offered a more holistic understanding of reservoir performance, supporting more informed decisions in well placement, injection strategy, and production planning (Rodriguez et al., 2024).

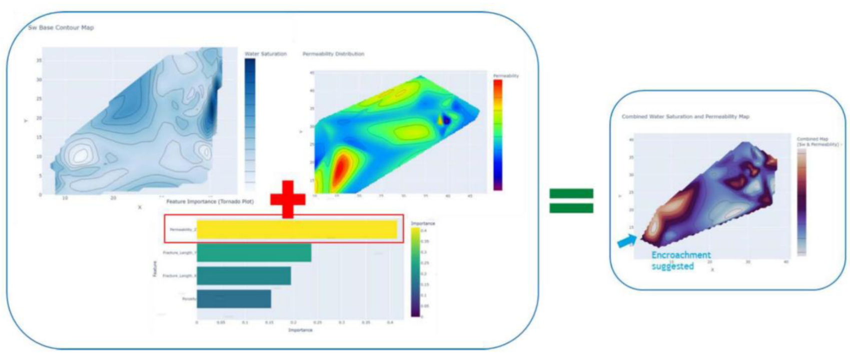


Figure 3—Combination of saturation maps with the dominant factors assessed from the machine learning model, new trend maps can be generated to analyze water saturation and its relationship with water influx

Collectively, these studies demonstrate that unsupervised learning is not only effective in exploratory data analysis but also serves as a foundation for hybrid workflows that integrate supervised learning and physics-based modeling. By enabling automated classification, clustering, and anomaly detection, unsupervised learning enhances the interpretability of complex datasets, supports real-time operations, and improves the efficiency of reservoir development strategies.

Advantages and Challenges

Advantages:

- No need for labeled data.
- Capable of discovering unknown patterns.

- Enhances geological and engineering interpretation.

Challenges:

- Sensitivity to feature scaling and selection.
- Difficulty in determining the optimal number of clusters.
- Interpretation of clusters requires domain expertise.

Future Directions

The integration of unsupervised clustering with real-time data streams, probabilistic modeling, and deep learning architectures (e.g., autoencoders, variational inference) is paving the way for more adaptive and intelligent reservoir management systems. As data volumes grow and computational tools evolve, clustering will continue to play a pivotal role in unlocking subsurface insights and optimizing field development strategies.

Methodology

This study introduces a data-driven methodology for enhancing zonation in lower completions design using unsupervised machine learning clustering (UMLC). The approach is designed to process real-time well log data, reduce dimensionality, and apply clustering to identify productive and non-productive zones along lateral wellbores. The methodology is structured into four main stages:

1. Data Acquisition and Ingestion
2. Preprocessing: Resampling and Outlier Detection
3. Dimensionality Reduction Using UMAP
4. Clustering with K-Means and Zonation Interpretation

A hybrid machine learning workflow is applied, combining UMAP for dimensionality reduction, K-Means clustering, and LSTM neural networks to a horizontal well drilled in a Middle Eastern clastic reservoir. This well, previously analyzed using deterministic and inversion-based workflows, will now be re-evaluated using a fully data-driven approach to enhance zonation and support completions design.

The dataset includes 2,711 depth-indexed rows and 31 features derived from real-time and post-drilling measurements. These features include bottomhole pressure (BHP), temperature (TEMP), rate of penetration (ROP), net gas, water rate, flow rate, permeability estimates, and other drilling and formation evaluation parameters.

Data Preparation and Preprocessing

We will begin by importing the well log data using the lasio library and converting it into a structured DataFrame. The dataset will be cleaned by removing rows with missing values using dropna(). The depth column (MD) will be retained as the index to preserve spatial continuity.

The features will then be standardized using StandardScaler to ensure that all variables contribute equally to the clustering and neural network training. This step is essential for both UMAP and LSTM, which are sensitive to feature scaling.

Dimensionality Reduction with UMAP

We will apply Uniform Manifold Approximation and Projection (UMAP) to reduce the high-dimensional feature space to two components. This transformation will allow us to visualize the data structure and improve the performance of the clustering algorithm by preserving both local and global relationships in the data.

Clustering with K-Means. We will perform K-Means clustering on the UMAP-reduced data. The optimal number of clusters will be determined by evaluating the silhouette score across a range of cluster values. The value of k that yields the highest silhouette score will be selected as the optimal number of clusters.

Once the optimal k is identified, we will re-run K-Means with this value and assign cluster labels to each depth interval. These labels will be added to the original DataFrame as a new column named Cluster.

LSTM Model for Cluster Prediction. To enhance the temporal consistency of the clustering and capture sequential dependencies in the data, we will train a Long Short-Term Memory (LSTM) neural network. The steps will include:

- Encoding the cluster labels using LabelEncoder.
- Splitting the dataset into training and testing sets (70/30 split).
- Reshaping the input data into 3D format: [samples, time steps, features].
- Converting the target labels to categorical format using to_categorical.

The LSTM model will be built using Keras with the following architecture:

- One LSTM layer with 50 units and ReLU activation.
- A dropout layer with a rate of 0.2 to prevent overfitting.
- A dense layer with 50 units and ReLU activation.
- An output layer with softmax activation for multi-class classification.

The model will be compiled with the Adam optimizer and categorical cross-entropy loss and trained for 50 epochs with a batch size of 32.

Cluster Refinement and Evaluation. After training, the LSTM model will be used to predict cluster labels for both training and testing sets. These predictions will be used to update the Cluster column in the DataFrame, effectively refining the initial K-Means clustering based on temporal learning.

We will evaluate the model using:

- Accuracy on the test set.
- Confusion matrix to assess classification performance.
- Training history plots for accuracy and loss over epochs.

Zonation and Export. We will generate zonation intervals by grouping consecutive depth rows with the same cluster label. Each zone will be defined by a start and end depth and assigned a cluster classification. The resulting zonation table will be exported to a CSV file (Cluster_facies.csv) and a LAS file (df.las) for integration into reservoir modeling and completions planning.

Feature Importance and Interpretation. To interpret the clustering results, we will apply a Random Forest classifier to the final dataset and compute feature importances. The top features contributing to cluster differentiation will be visualized using bar plots and heatmaps. Additionally, we will perform Recursive Feature Elimination (RFE) with linear regression to identify the most influential features for zonation.

Table 2—Original Workflow

Component	Description
Data Sources	LWD logs (GR, Rt, Rh, Rv, density, neutron), EDAR, ABDI
Preprocessing	Manual interpretation, structural inversion (HAHZ), quick-look petrophysics
Zonation Method	Stratigraphic Modified Lorenz Plot (SMLP) and Modified Lorenz Plot (MLP) based on permeability index
Zonation Criteria	Manual thresholds on porosity, resistivity, and GR; EDAR-derived thickness and saturation
Output	Ranked flow/storage zones; ICD and packer placement recommendations
Limitations	Subjective interpretation, limited automation, no temporal modeling, no clustering

This workflow relied on deterministic calculations and interpreter-defined thresholds to classify zones. While effective, it required significant manual effort and was sensitive to interpreter bias. The use of EDAR and ABDI improved structural understanding, but the zonation process remained static and univariate in nature.

Table 3—UMLC Workflow (UMAP + KMeans + LSTM)

Component	Description
Data Sources	LWD logs, including Ultra-Deep Azimuthal Resistivity data & Permeability calculation from customer.
Preprocessing	Automated resampling and outlier removal; no manual smoothing or imputation
Dimensionality Reduction	UMAP to preserve local/global structure and improve clustering
Clustering Method	KMeans clustering on UMAP-reduced data; optimal k selected via silhouette score
Temporal Modeling	LSTM neural network trained on sequential data to refine cluster assignments
Zonation Output	Cluster-based zonation intervals with start/end depths and cluster labels
Validation	Accuracy, confusion matrix, feature importance (Random Forest), RFE, correlation analysis
Export	Zonation exported to CSV and LAS; visualizations include UMAP plots, feature importance, and pair plots

The UMLC approach provided a more granular and objective classification of reservoir zones compared to traditional methods. The clustering results aligned well with known geological features and production data, confirming the validity of the approach.

Table 4—Comparison of User defined workflow vs UMLC workflow

	Original Workflow	UMLC Workflow
Zonation Accuracy	Based on permeability index and interpreter judgment	Based on multidimensional clustering and LSTM refinement
Resolution	Limited by manual thresholds and visual inspection	Enhanced by UMAP projection and LSTM sequence modeling
Automation	Low	High
Bias Sensitivity	High (interpreter-dependent)	Low (data-driven)
Temporal Consistency	Not modeled	Explicitly modeled via LSTM
Feature Utilization	Limited to a few logs	Full use of 30+ features including operational metrics
Output Format	Static plots and tables	Dynamic visualizations, CSV, LAS, and model diagnostics

This comparative framework sets the stage for the application of the new workflow. In the next section, you will present the results of the UMAP + KMeans + LSTM analysis, including zonation maps, model performance metrics, and feature importance rankings.

Results

In this section, we will discuss the user-defined analysis, including zonation results and properties, comparing it to UMLC Zonation results.

Zonation Based on Integrated User-Defined Analysis

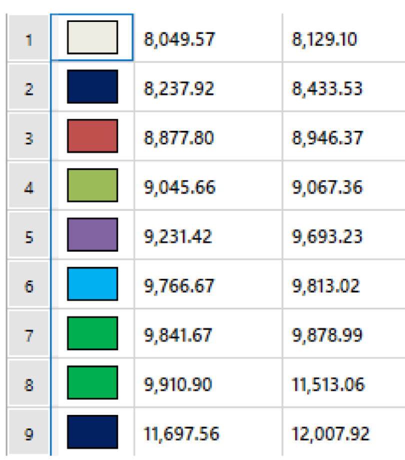
Following the application of the proposed multi-disciplinary workflow, a final zonation of the reservoir was established. This zonation integrates structural and petrophysical insights derived from Logging While Drilling (LWD) measurements, Extra-Deep Azimuthal Resistivity (EDAR) inversions, and image-constrained resistivity modeling (HAHZ structural inversions). The objective was to delineate productive and non-productive intervals along the horizontal section of a well drilled in a Middle Eastern channel sand reservoir.

The zonation process incorporated the following key inputs:

- **True formation resistivity (Rt)** derived from HAHZ structural inversions, minimizing artefacts such as polarization horns.
- **Permeability index** calculated from quick-look petrophysical evaluation using multiple empirical models.
- **Porosity and hydrocarbon saturation** derived from conventional LWD logs.
- **Pay zone thickness (PZT)** obtained from EDAR inversions.
- **Flow and storage capacity** visualized through Stratigraphic Modified Lorenz Plots (SMLP) and Modified Lorenz Plots (MLP).
- **Borehole shape and trajectory data**, including caliper logs and dogleg severity (DLS), to assess completion feasibility.

Zonation Outcome

The final ranking of reservoir zones, from highest to lowest expected productivity, is as follows: Zone 3, 5, 6, 9, 7, 1, 2, 8, 4. Zones length are disclosed in figure X below



1		8,049.57	8,129.10
2		8,237.92	8,433.53
3		8,877.80	8,946.37
4		9,045.66	9,067.36
5		9,231.42	9,693.23
6		9,766.67	9,813.02
7		9,841.67	9,878.99
8		9,910.90	11,513.06
9		11,697.56	12,007.92

Figure 4—User-defined Zones length.

This ranking reflects a comprehensive evaluation of both reservoir quality and operational constraints. Notably, **Zone 3** was identified as the most favorable interval, exhibiting a combination of high permeability, favorable saturation, and consistent thickness. **Zones 5 and 6** also demonstrated strong reservoir characteristics, while **Zone 9**, initially considered marginal, was upgraded due to its improved storage profile after incorporating EDAR-derived thickness and saturation data.

Conversely, **Zones 1 and 2**, which may have been prioritized under conventional evaluation methods, were downgraded due to lower permeability and the presence of high dogleg severity, which could complicate completion operations. **Zone 4** consistently ranked lowest across all evaluation criteria, indicating limited production potential. Results can be observed in figure X below:

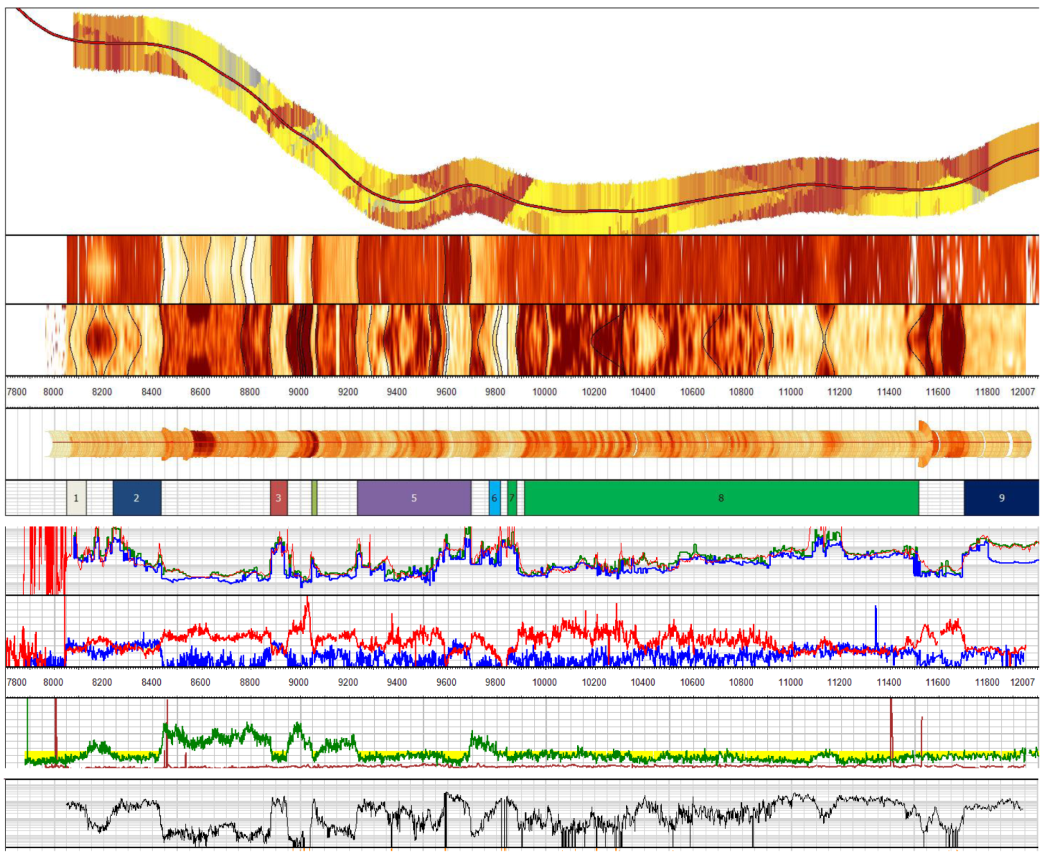


Figure 5—User-Defined zone analysis vs. features. From top to bottom: Dips Model from Borehole Density Image dip picks; Gamma Ray Image, Density Image, 3D caliper, User-defined Zones, LWD Resistivity curves, LWD bulk Density/Neutron curves, LWD Bulk Gamma Ray curves with Average ROP, Calculated Permeability from customer.

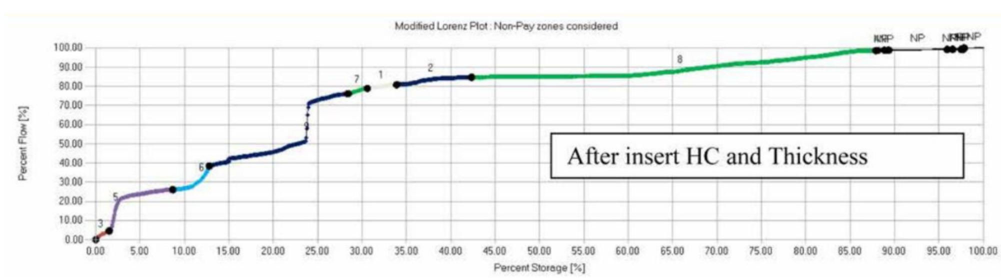


Figure 6—Modified Lorenz plot showing the highest to lowest flow vs storage capability per region.

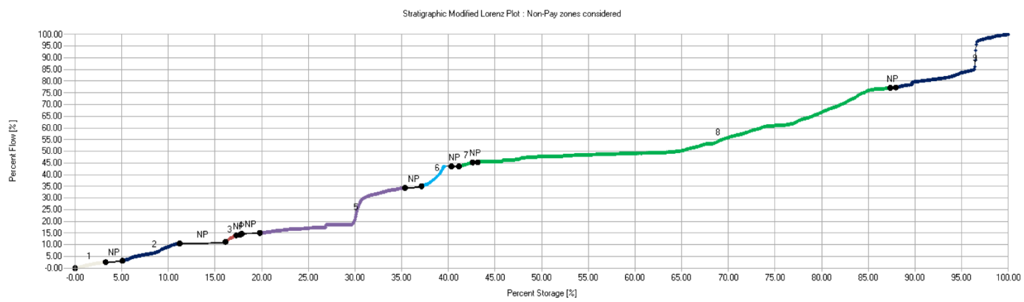


Figure 7—Stratigraphic Lorenz plot showing the highest to lowest flow vs storage capability per region.

Zonation Analysis from Unsupervised Learning

A detailed visual inspection of the cluster-based zonation derived from unsupervised learning applied to LWD and azimuthal resistivity data reveals a stratified and compartmentalized reservoir architecture. The clusters, labeled numerically from 1 to 7, are distributed heterogeneously along the lateral section, each representing distinct petrophysical facies.

Cluster Distribution and Petrophysical Interpretation

The clusters exhibit varying degrees of lateral continuity and petrophysical quality, summarized as follows:

- **Cluster 6**: This cluster is the most favorable, consistently associated with intervals of high resistivity and moderate-to-high porosity. These characteristics suggest clean, hydrocarbon-bearing sands with excellent reservoir quality.
- **Cluster 2**: Slightly more heterogeneous than Cluster 6, this cluster still demonstrates favorable log responses and is interpreted as a high-quality reservoir facies.
- **Cluster 0**: Represents a transitional facies with moderate resistivity and porosity. It may indicate partially clean sands or mixed lithologies with variable saturation.
- **Cluster 4**: Exhibits moderate petrophysical quality and appears in zones with mixed log responses, suggesting heterogeneity or transitional depositional environments.
- **Clusters 7 and 5**: These clusters are associated with lower resistivity and porosity, and higher water saturation. They likely represent lower-quality reservoir intervals or water-bearing sands.
- **Cluster 3**: Characterized by tight, resistive intervals with low porosity, this cluster is interpreted as non-pay or as a potential baffle to fluid flow.
- **Cluster 1**: The least favorable cluster, consistently associated with poor petrophysical properties, including low resistivity, low porosity, and high water saturation. It likely corresponds to non-reservoir or shale-dominated intervals.

Reservoir Compartmentalization and Operational Implications

The alternating pattern of clusters along the lateral section indicates a high degree of reservoir compartmentalization. Laterally continuous clusters such as 6 and 2 suggest the presence of extensive, high-quality reservoir zones, while the more fragmented appearance of clusters like 3 and 1 may indicate localized barriers or non-reservoir facies.

This zonation provides a robust framework for operational decision-making. High-quality clusters (6, 2, 0) are prime candidates for completion and stimulation, whereas low-quality or tight clusters (3, 1) may be targeted for mechanical isolation (e.g., packer placement). The clustering results thus offer a data-driven basis for optimizing well completion strategies and enhancing reservoir management.

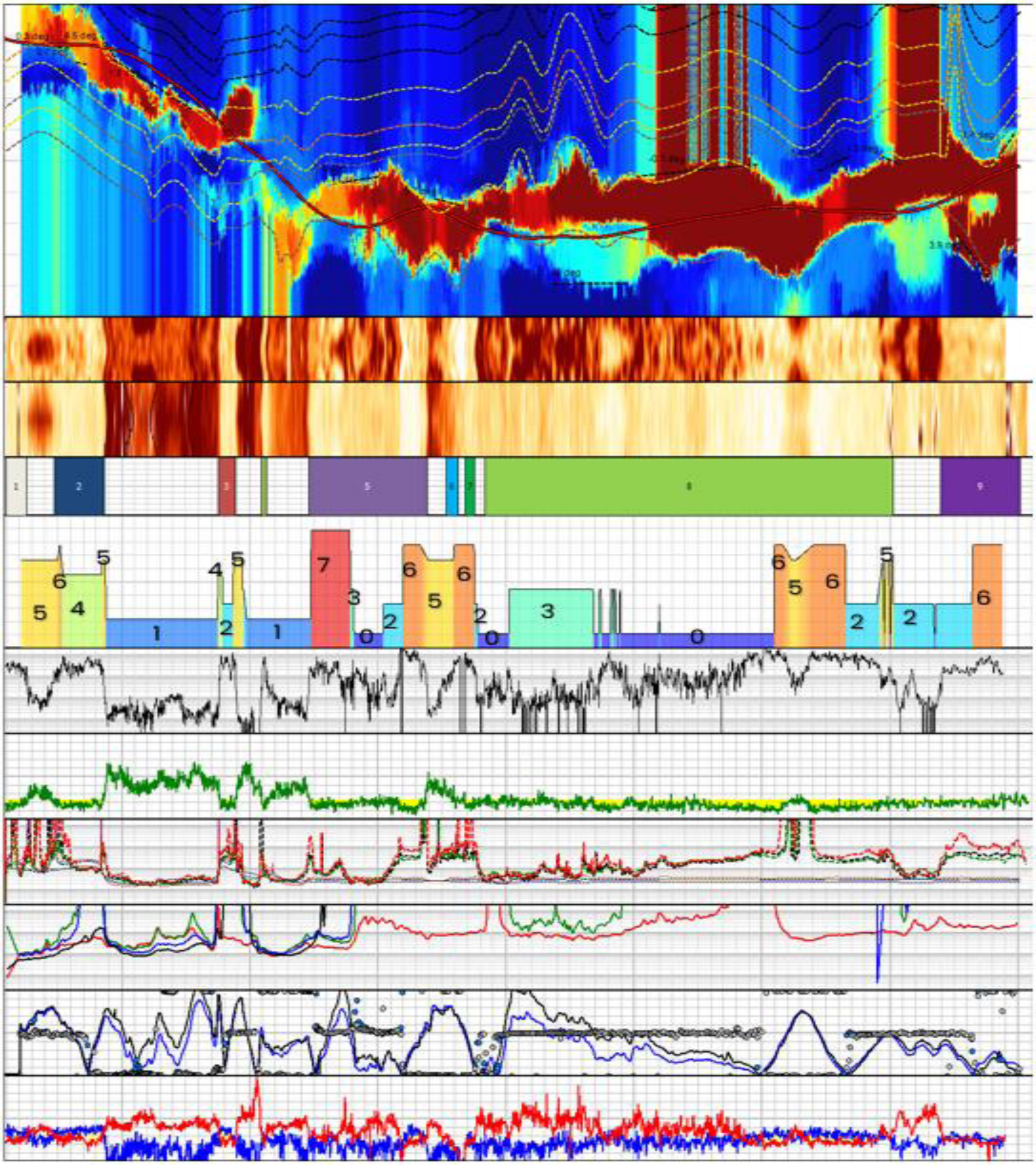


Figure 8—UMLC zones analysis vs. features. From top to bottom: Inversions Results; Gamma Ray Image, Density Image, User-defined zones; UMLC Zones, LWD Bulk Gamma Ray curve, LWD Resistivity curves, LWD UDAR Resistivity Curves, LWD UDAR Azimuthal Signal Strength & Target Angle; LWD bulk Density/Neutron curves.

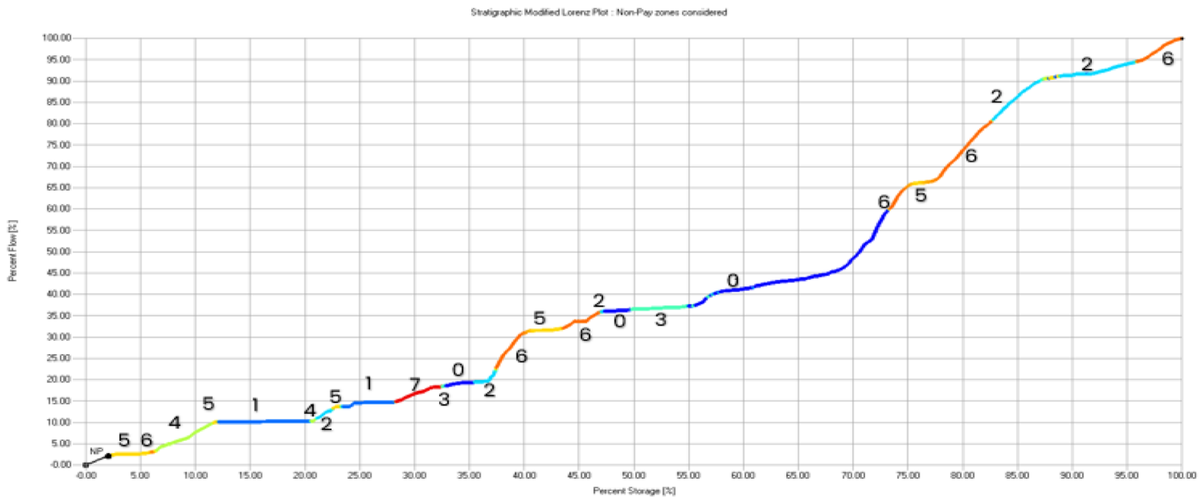


Figure 9—Stratigraphic Lorenz plot showing the highest to lowest flow vs storage capability per region based on UMLCS.

Confidence Analysis of Unsupervised Machine Learning Clustering (UMLC)

The application of unsupervised machine learning clustering (UMLC) to reservoir characterization has gained traction due to its ability to uncover latent patterns in high-dimensional petrophysical data without requiring labeled inputs. However, the reliability of such clustering outcomes depends critically on the robustness of the feature extraction process, the quality of the clustering solution, and the interpretability of the resulting clusters.

To address these concerns, this study incorporates a comprehensive confidence analysis framework into the UMLC workflow. This framework includes model performance evaluation, cluster validation metrics, dimensionality reduction for visual inspection, and intra-cluster variability analysis. Together, these diagnostics provide a quantitative and qualitative basis for assessing the trustworthiness of the clustering results and their applicability to reservoir zonation and completion design.

Model Accuracy and Loss. These plots evaluate the performance of the neural network used for feature extraction (e.g., an autoencoder). The training and validation accuracy curves demonstrate convergence, while the loss curves confirm that the model avoids overfitting. This ensures that the extracted features are generalizable and suitable for clustering.

Silhouette Score vs. Number of Clusters. This plot assesses the quality of clustering for different values of k*k*. The silhouette score measures how similar an object is to its own cluster compared to other clusters. A peak in the silhouette score indicates the optimal number of clusters, supporting the selection of k=8 in this study.

UMAP Projection with kMeans Clustering. UMAP (Uniform Manifold Approximation and Projection) is used to reduce the high-dimensional feature space to two dimensions for visualization. The resulting plot shows well-separated clusters, confirming that the clustering algorithm has identified distinct facies.

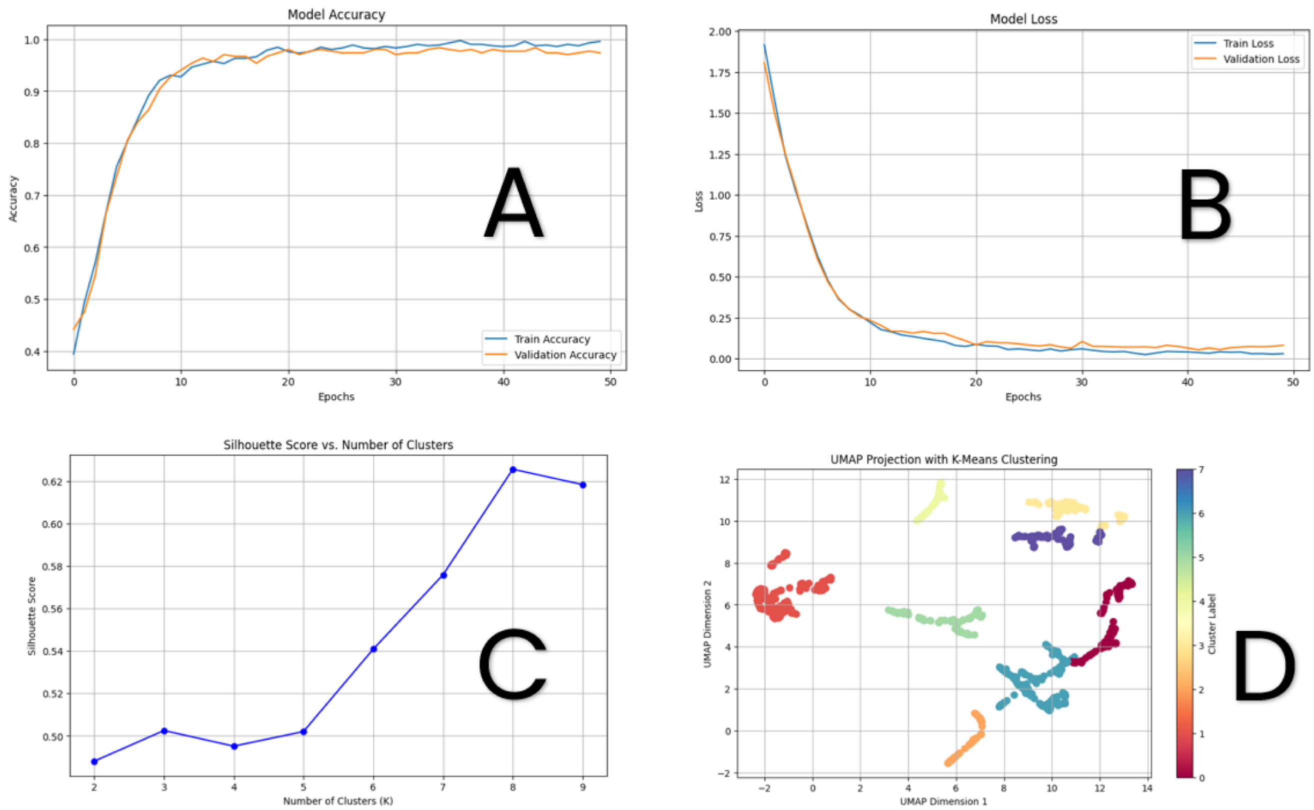


Figure 10—A. Training and validation accuracy over epochs for the feature extraction model. The convergence of both curves indicates stable learning and generalization, B. Training and validation loss over epochs. The decreasing and stabilizing loss values confirm effective model optimization. C. Silhouette score as a function of the number of clusters. The peak score identifies the optimal cluster count, balancing cohesion and separation. D. ***Two-dimensional UMAP projection of the feature space, colored by kMeans cluster assignment. The clear separation between clusters supports the validity of the clustering solution*****.**

Standard Deviation of Features in Each Cluster. This bar chart quantifies the internal variability of each cluster. Clusters with low standard deviation are more homogeneous and thus more reliable for operational decisions. Higher variability may indicate transitional zones or mixed lithologies.

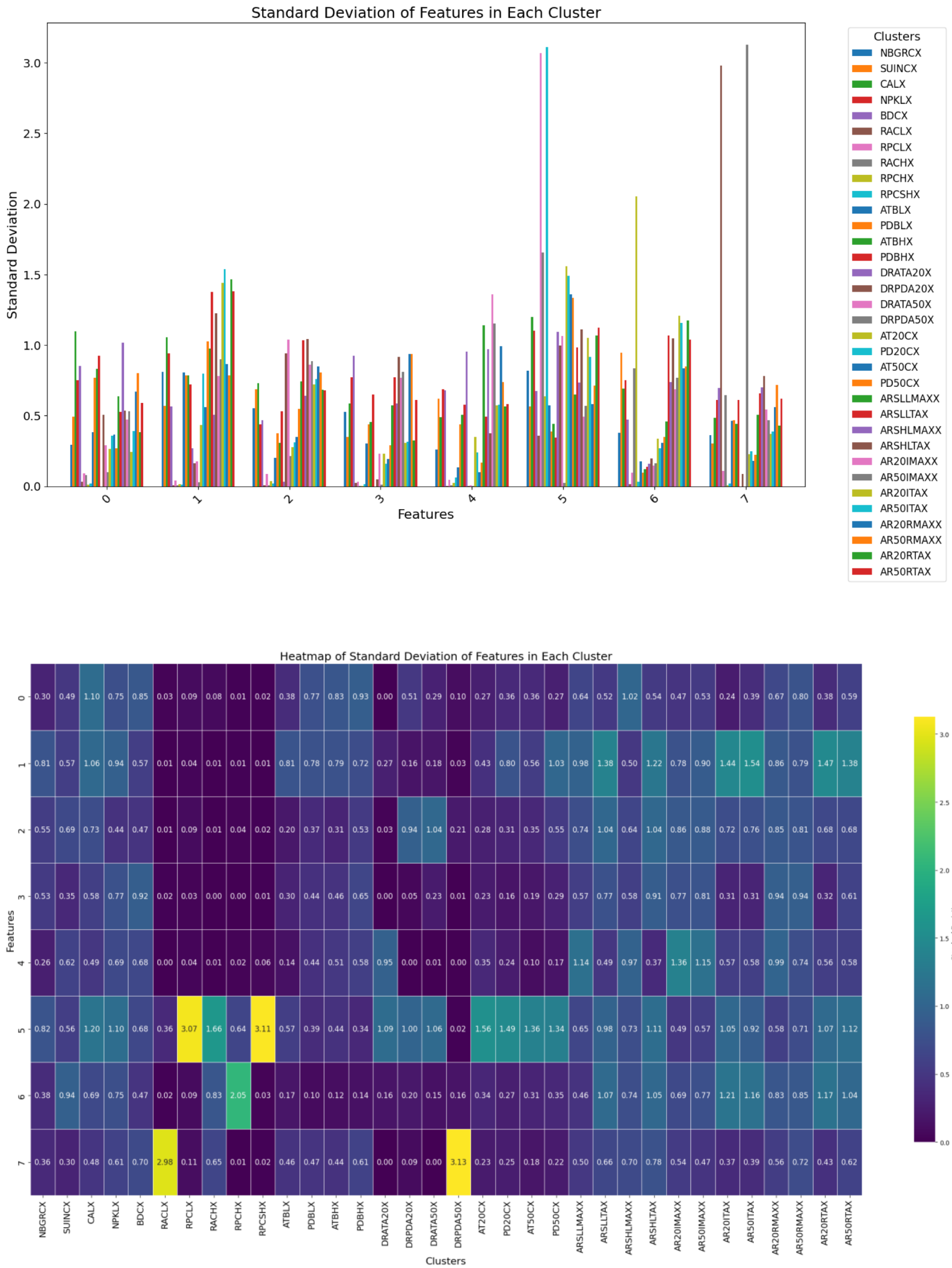


Figure 11—Standard deviation of input features within each cluster. Lower values indicate more homogeneous clusters, which are preferable for zonation and completion planning.

Feature Importance and Decision Logic in Clustering Interpretation

To enhance the interpretability of the unsupervised clustering results, a post hoc analysis was conducted using feature importance ranking and decision tree visualization. These tools provide insight into the relative influence of input features on the clustering outcome and offer a transparent view of the decision logic that underpins cluster differentiation.

Feature Importance Analysis. The bar chart titled *"Top 5 Most Important Features"* presents the relative importance of input variables in determining cluster membership. The feature importance values were likely derived from a tree-based model (e.g., Random Forest or Gradient Boosted Trees) trained to predict cluster labels assigned by the unsupervised algorithm.

This ranking confirms that clustering was not dominated by a single variable but rather reflected a balanced integration of multiple features.

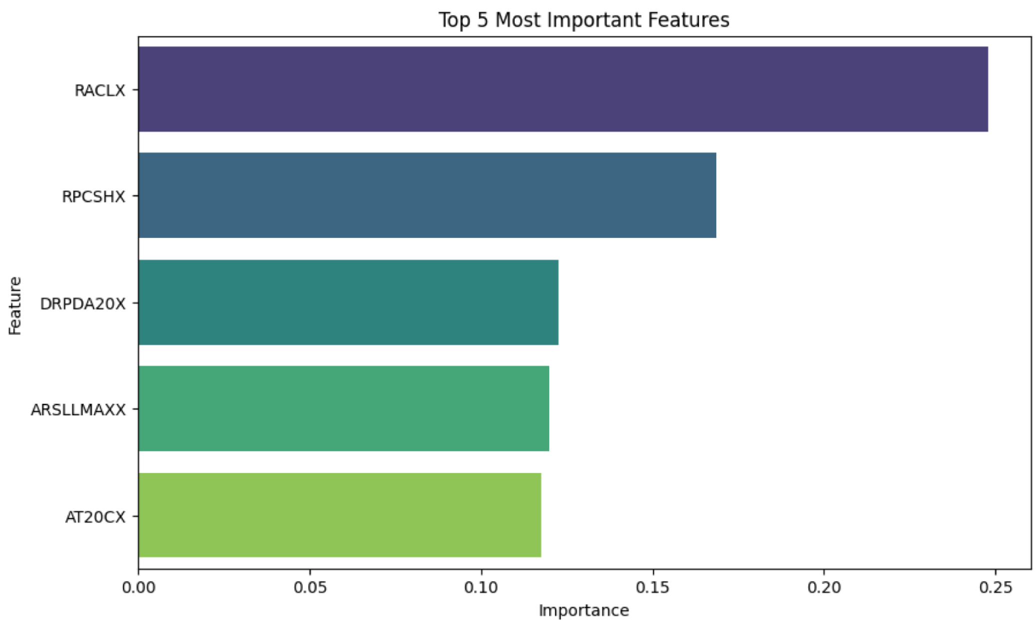


Figure 12—Bar chart showing the top five most important features influencing cluster assignment. Feature importance was derived from a supervised model trained on unsupervised cluster labels.

Decision Tree Visualization. The decision tree diagram provides a hierarchical breakdown of how feature thresholds contribute to cluster classification. Each node represents a decision based on a specific feature value, and each branch leads to a subsequent decision or final cluster assignment.

This visualization serves two purposes:

1. **Interpretability**: It translates the abstract clustering results into a rule-based structure that domain experts can understand and validate.
2. **Operational Utility**: The tree can be used as a lightweight decision-support tool for assigning new data points to clusters in real time.

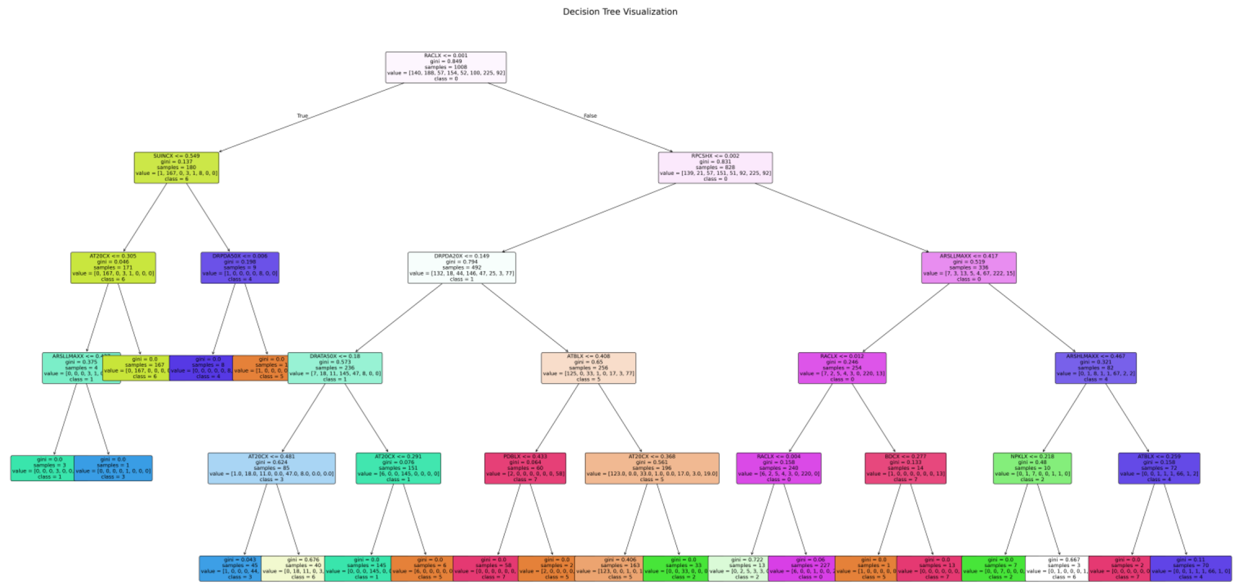


Figure 13—Decision tree visualization illustrating the logic used to classify data points into clusters based on feature thresholds. The tree structure enhances interpretability and supports real-time application.

Conclusions

This study demonstrates the efficacy of unsupervised machine learning clustering (UMLC) as a robust tool for enhancing reservoir zonation in support of lower completions design. By leveraging real-time logging-while-drilling (LWD) measurements and advanced clustering algorithms, the proposed methodology enables a more objective and data-driven classification of reservoir intervals, independent of permeability input.

The key advantages of this approach include:

- **Enhanced Zonation Resolution**: The clustering framework improves the detection of subtle petrophysical heterogeneities, enabling more refined reservoir compartmentalization.
- **Operational Efficiency**: The automated nature of the workflow reduces interpretation time and supports more informed decisions regarding inflow control device (ICD) and packer placement.
- **Scalability and Adaptability**: The methodology is applicable across multiple wells and can be adapted to a variety of lithological and depositional settings.

The integration of UMLC into traditional petrophysical workflows represents a significant advancement in data-driven reservoir characterization and completion planning.

Future Work

To further validate and expand the applicability of the proposed methodology, future research will focus on the following areas:

- **Field-Scale Implementation**: Deployment of the clustering workflow across a broader portfolio of wells to evaluate its scalability, robustness, and consistency in diverse geological settings.
- **Real-Time Integration**: Development of a real-time clustering module for deployment during drilling operations, enabling dynamic decision-making for zonal isolation and completion design.
- **Hybrid Modeling Approaches**: Integration of unsupervised clustering with supervised learning techniques to enhance predictive capabilities in wells lacking comprehensive log suites.
- **Cross-Disciplinary Integration**: Embedding clustering outputs into collaborative platforms to facilitate joint interpretation by geologists, petrophysicists, and completions engineers.

Additionally, the methodology holds promise for application in renewable energy domains, such as geothermal reservoirs, where accurate zonation is critical for optimizing sustainable production strategies.

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